Message-Level Communication Functions in LLM Multi-Agent Systems: A Taxonomy and Empirical Study

Anonymous Author(s)
Affiliation
Address
email

Abstract

Removing the least informative messages from LLM multi-agent conversations produces the largest behavioral disruption—a finding that, while well-established for human backchannels in conversation analysis, has never been quantified in LLM-based multi-agent systems (MAS). Across 2,693 GSM8K traces spanning three 7–9B backbones and three communication topologies, perturbation-based ablation reveals that excising simple acknowledgments changes downstream behavior far more (mean similarity $= 0.35$) than removing content-rich reasoning guidance (0.78); a single removed acknowledgment alters the final answer in 66% of cases versus 29% for reasoning guidance. We trace this pattern to three backbone-dependent communication modes: *function-driven* (Qwen; $F = 232$, $p < 0.001$), *position-driven* (Gemma; $R^2 = 0.34$, function type n.s.), and *minimally differentiated* (Llama; $R^2 = 0.02$). We additionally report a preliminary crossover interaction suggesting that dynamic routing modulates these modes. These results provide the first quantitative evidence that the structural load-bearing role of minimal responses, long documented in human conversation, extends to LLM agents—and that this structure is model-dependent rather than universal.

1 Introduction

Communication optimization in LLM-based multi-agent systems (MAS) rests on an implicit assumption: message value correlates with information content. Pruning methods Wang et al. [2025b], Li et al. [2025], Zhang et al. [2025a] discard messages deemed redundant or low-content; routing methods Li et al. [2026], Chen et al. [2026a] direct information-rich messages to relevant agents. In human conversation, decades of conversation analysis (CA) research have established the opposite: backchannels Yngve [1970], continuers Schegloff [1982], and acknowledgment tokens Jefferson [1984] are structurally load-bearing despite carrying little propositional content Clark and Schaefer [1989]. Whether this extends to LLM agents—which lack human social cognition—has never been empirically tested.

We provide the first quantitative evidence that it does, while revealing backbone-dependent structure invisible to single-model analyses. Through perturbation-based ablation of 2,693 multi-agent GSM8K traces across three backbones and three topologies, we show that removing acknowledgments produces the *largest* behavioral disruption (mean similarity $= 0.35$), while removing reasoning guidance produces the *smallest* (0.78). A single excised acknowledgment alters the final answer in 66% of cases versus 29% for reasoning guidance—demonstrating that content-based pruning heuristics Wang et al. [2025b], Liang et al. [2025] would discard the wrong messages.

We identify two primary contributions and an exploratory finding:

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1. **Structural load-bearing of minimal responses** (C1). The per-type impact gradient is inversely related to information content, providing the first quantitative verification of the classical CA finding in LLM MAS. A neutralize ablation—preserving structure while removing content—produces the opposite gradient, confirming that acknowledgments' high impact stems from structural presence (Appendix E).
2. **Three backbone-dependent communication modes** (C2). OLS ANCOVA reveals the impact gradient is backbone-dependent: Qwen exhibits *function-driven* communication, Gemma *position-driven*, and Llama *minimally differentiated*. C1 is the instantiation of C2's function-driven mode on Qwen.

We additionally report a *preliminary* crossover interaction: selector topology partially converges the function-driven and minimally differentiated modes ($\eta^2 = 0.043$), though mixed-effects analysis indicates this is strongly modulated by task-level clustering (ICC $= 0.73$). Cross-domain validation on MATH provides preliminary support; temperature robustness checks confirm stable results (Appendices H, C).

Our approach combines a five-category functional taxonomy validated via cross-model annotation ($\kappa = 0.35$–0.67) with perturbation-based ablation isolating each message's behavioral impact. Foundational work Sukhbaatar et al. [2016], Foerster et al. [2016] establishes *how* agents learn to communicate; theoretical work Rizvi-Martel et al. [2026] establishes *when* communication helps; trace-level diagnosis Cemri et al. [2025] identifies *what went wrong*. We provide the missing *what*: a functional characterization suggesting communication structure in LLM MAS is model-dependent rather than universal, at least within the 7–9B scale and GSM8K domain studied here.

2 Related Work

Our work draws on conversation analysis, multi-agent communication, and message function analysis.

Conversation analysis and grounding. Backchannels Yngve [1970], continuers Schegloff [1982], and acknowledgment tokens Jefferson [1984] are structurally load-bearing in human conversation; Clark and Schaefer Clark and Schaefer [1989] formalized grounding as requiring evidence of understanding. We test whether this transfers to LLM agents.

Multi-agent communication. CommNet Sukhbaatar et al. [2016] and DIAL Foerster et al. [2016] showed agents can learn communication end-to-end; IMAC Wang et al. [2020] optimized bandwidth via information bottleneck. Rizvi-Martel *et al.* Rizvi-Martel et al. [2026] derive bounds on *when* communication helps; Chen *et al.* Chen et al. [2026b] identify "what to communicate" as understudied. LLM frameworks (CAMEL Li et al. [2023], MetaGPT Hong et al. [2024], AG2 Wu et al. [2024], AgentVerse Chen et al. [2024]) demonstrate communication affects performance; debate Du et al. [2024] improves reasoning but may reduce to majority voting Choi et al. [2025], and intrinsic strength dominates structure Wu et al. [2025], de Font-Reaulx et al. [2026]. Yan *et al.* Yan et al. [2025] identify message-level analysis as an open direction.

Communication efficiency. AgentDropout Wang et al. [2025b], AgentPrune Zhang et al. [2025a], AGP Li et al. [2025], and OPTIMA Chen et al. [2025] achieve 20–90% token reduction but treat message utility as scalar. S^2-MAD Liang et al. [2025] and SHARP Kim et al. [2026] benchmark debate and role harmonization at scale. Zhang *et al.* Zhang et al. [2025b] analyze dialogue acts in collaborative LLM agents; we differ by measuring perturbation impact via ablation, providing ANCOVA quantification, and discovering a backbone $\times$ topology interaction.

Attribution and pragmatics. AgentSHAP Horovicz et al. [2025] and MAST Cemri et al. [2025] operate at agent/trace level; our ablation provides message-level granularity. Classical speech act theory Austin [1962], Searle [1969] and dialogue act taxonomies Core and Allen [1997], Bunt et al. [2012] inform our categories. LLM prompt sensitivity Sclar et al. [2024], He et al. [2024] and pragmatic models Estienne et al. [2025], Wang et al. [2025a] provide complementary lenses.

Table 1: Taxonomy of communication function types with definitions and examples from GSM8K solver–checker dialogues.

Function Type	Definition	Example
INFORMATION PROVISION	Supplies factual content, data, or intermediate results	"The total cost is 3 × $5 = $15 for the shirts."
REASONING GUIDANCE	Directs reasoning through decomposition or strategic suggestions	"First calculate the weekly cost, then multiply by 4."
VERIFICATION	Checks or corrects the correctness of prior messages	"17 × 3 = 51, not 53. The corrected answer is $51."
COORDINATION	Manages dialogue flow, assigns roles, or delegates tasks	"I'll solve first, then you verify step by step."
ACKNOWLEDGMENT	Simple confirmations or agreements with no new content	"I agree with your solution. The answer is 42."

3 Taxonomy of Communication Functions

We define five exhaustive, mutually exclusive communication function types (Table 1). Ambiguity is resolved by hierarchical priority: *coordination* $\succ$ *verification* $\succ$ *reasoning_guidance* $\succ$ *information_provision* $\succ$ *acknowledgment*, reflecting the principle that structurally higher-level functions take precedence.

INFORMATION PROVISION supplies facts that expand the shared state without directing how they should be used. REASONING GUIDANCE is prescriptive: it tells the recipient *how* to proceed. VERIFICATION assesses prior contributions' correctness. COORDINATION manages meta-structure (roles, turn-taking) without contributing task content. ACKNOWLEDGMENT confirms receipt or agreement; while informationally sparse, it maintains conversational coherence.

Our categories map onto established dialogue act frameworks (DAMSL Core and Allen [1997], ISO 24617-2 Bunt et al. [2012], FIPA ACL Foundation for Intelligent Physical Agents [2002]; detailed mapping in Appendix L), collapsing 56+ fine-grained types into five categories operationalized via measurable perturbation impact rather than speaker intent.

Annotation protocol. Each message is annotated by an LLM annotator using a structured prompt containing taxonomy definitions, decision rules, and three in-context examples per type, with temperature 0 and constrained output. Cross-model validation uses two independent annotator models per message; we adopt $\kappa \geq 0.4$ as our target threshold for cross-model agreement and report Cohen's κ throughout (Section 6.4).

4 Methodology

4.1 Multi-Agent System Setup

We use AG2 v0.6.1 (AutoGen) Wu et al. [2024] with solver–checker agent teams across three backbones of comparable scale: Qwen2.5-7B-Instruct Yang et al. [2024], Llama-3.1-8B-Instruct Grattafiori et al. [2024], and Gemma-2-9B-IT Gemma Team [2024] (7–9B parameters). We study three topologies: **round-robin** (fixed cyclic order), **sequential** (strict two-phase rounds), and **selector** (dynamic routing via a selector agent). All models are served via vLLM v0.6.6 Kwon et al. [2023] with greedy decoding (temperature $= 0$, max tokens $= 2048$). For each of the $3 \times 3 = 9$ configurations, we collect traces on 300 GSM8K Cobbe et al. [2021] test-split tasks sampled with seed $= 42$, yielding 2,693 non-empty traces. The MATH cross-domain validation (Appendix H) uses 50 tasks at Level 3–5 (Algebra and Number Theory), also sampled with seed $= 42$. The full annotation prompt is reproduced in Appendix I.

4.2 Message-Level Ablation

To isolate each message's perturbation effect on downstream behavior, we introduce two ablation types. **Remove ablation** deletes message m_i from trace $\mathcal{T}$ and regenerates m_{i+1}. **Neutralize ablation** replaces m_i with "`Acknowledged.`", preserving positional structure while removing content. Comparing the two disentangles content effects from structural presence effects McNemar [1947].

We measure impact as:

$$\text{sim}(m_i) = \cos\big(\mathbf{e}(m_{i+1}),\, \mathbf{e}(m'_{i+1})\big) \tag{1}$$

where $\mathbf{e}(\cdot)$ is a sentence embedding (`all-MiniLM-L6-v2`; Reimers and Gurevych [2019]). Lower similarity indicates higher message impact. We report similarity throughout; impact $= 1 - \text{sim}$.

4.3 Statistical Framework

We test whether function types produce different impact distributions using the Kruskal–Wallis H test Kruskal and Wallis [1952] (non-parametric, appropriate for bounded, skewed distributions), with Dunn's post-hoc tests Dunn [1964] and Bonferroni correction.

To control for the position confound—function types are non-uniformly distributed across conversation positions—we fit an OLS ANCOVA:

$$\text{sim}(m_i) = \beta_0 + \sum_{j=1}^{4} \beta_j \cdot \mathbb{1}[\text{type}(m_i) = j] + \gamma_1 \cdot \text{pos}(m_i) + \gamma_2 \cdot \text{round}(m_i) + \epsilon_i \tag{2}$$

The partial F-statistic for the function type factor tests whether function type effects persist after controlling for position and round number. We report HC3 heteroscedasticity-consistent standard errors throughout; residual diagnostics are in Appendix D. Because we test function type significance across $3 \times 3 = 9$ configurations, we report Benjamini–Hochberg (BH) adjusted p-values to control the false discovery rate. We further validate with a pooled three-way ANOVA across all backbones and topologies (Section 6.4).

5 Experiments

Each agent team uses AG2's `ConversableAgent` with task-specific system prompts (solver: step-by-step solutions; checker: verify and flag errors). Under selector topology, a third agent routes messages via `GroupChat` with `speaker_selection_method="auto"`. Conversations are limited to 10 rounds. Function type labels are assigned by two independent LLM annotators (Qwen2.5-7B as Model A, Llama-3.1-8B as Model B) with structured JSON output. Solution correctness is determined by comparing extracted numerical answers against GSM8K ground truth.

6 Results

We report results across all nine backbone–topology configurations.

6.1 Impact Gradient

For Qwen under fixed-turn topologies (Figure 1), removing ACKNOWLEDGMENT messages—the least informative category—produces the largest behavioral change (mean similarity $= 0.352$, $n = 450$), followed by INFORMATION_PROVISION (0.410, $n = 177$), VERIFICATION (0.616, $n = 110$), and REASONING_GUIDANCE (0.778, $n = 158$). The impact gradient is inversely related to information content: informationally sparse acknowledgments are more impactful than content-rich reasoning guidance.

Pruning evidence. Individual-message ablation confirms this gradient at the single-message level: removing one acknowledgment alters the final answer in 66% of cases, versus 29% for reasoning guidance. The neutralize ablation—which preserves message structure while removing content—produces the opposite gradient: information-rich types suffer most. This double dissociation confirms that acknowledgment's high remove-impact stems from structural presence rather than informational

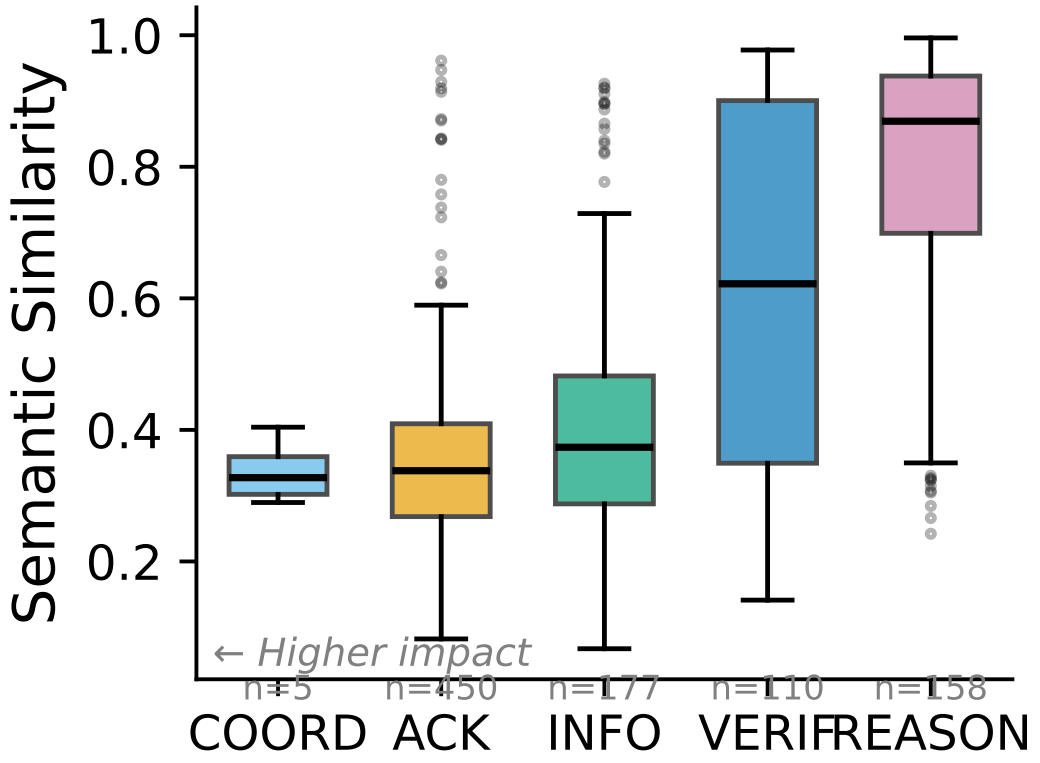


Figure 1: Remove-ablation semantic similarity (MiniLM cosine) by function type for Qwen2.5-7B under round-robin topology ($n = 895$). Lower similarity indicates greater behavioral disruption when the message is removed. Boxes show IQR; whiskers extend to $1.5\times$ IQR. Acknowledgment messages (median $= 0.35$) are the most impactful despite carrying the least information content.

Table 2: Kruskal–Wallis H statistics for semantic similarity across function types. All $p < 0.01$. Group counts vary (4–5 under RR/SEQ, 2–3 under selector); H values not directly comparable across topologies.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	287.4	288.1	50.8
Gemma-2-9B	21.5	25.1	6.9
Llama-3.1-8B	13.0	15.8	63.8

content (Appendix E). The gradient is robust across four independent text similarity metrics (Spearman $\rho > 0.91$; Appendix G) and across RR and SEQ topologies, but breaks down under selector topology.

6.2 Three Communication Modes

The Kruskal–Wallis H test confirms that function type significantly predicts message impact across all configurations ($p < 0.01$), but the magnitude varies by an order of magnitude across backbones (Table 2). However, this omnibus test conflates function type effects with positional confounds.

Position control via OLS ANCOVA. To disentangle function type from positional confounds, we fit the ANCOVA model (Equation 2). Table 3 reveals three qualitatively distinct communication modes that explain *why* the impact gradient exists for some backbones but not others.

Table 3: OLS ANCOVA: Type II partial F for function type (controlling for position and round) and full-model R^2. Significance stars reflect BH-adjusted p-values across 9 tests. Type II SS is used because function type and position are not orthogonal; Type III SS yields identical conclusions. $^{***}p_{\text{BH}} < 0.001$; $^{**}p_{\text{BH}} < 0.01$; $^{*}p_{\text{BH}} < 0.05$; n.s. = not significant.

	RR		SEQ		Selector	
Backbone	F	R^2	F	R^2	F	R^2
Qwen2.5-7B	232^{***}	.450	225^{***}	.446	55.8^{***}	.065
Gemma-2-9B	$0.43^{\text{n.s.}}$	.341	$0.44^{\text{n.s.}}$	.333	$0.60^{\text{n.s.}}$	.036
Llama-3.1-8B	3.81^{*}	.021	4.45^{**}	.020	17.3^{***}	.056

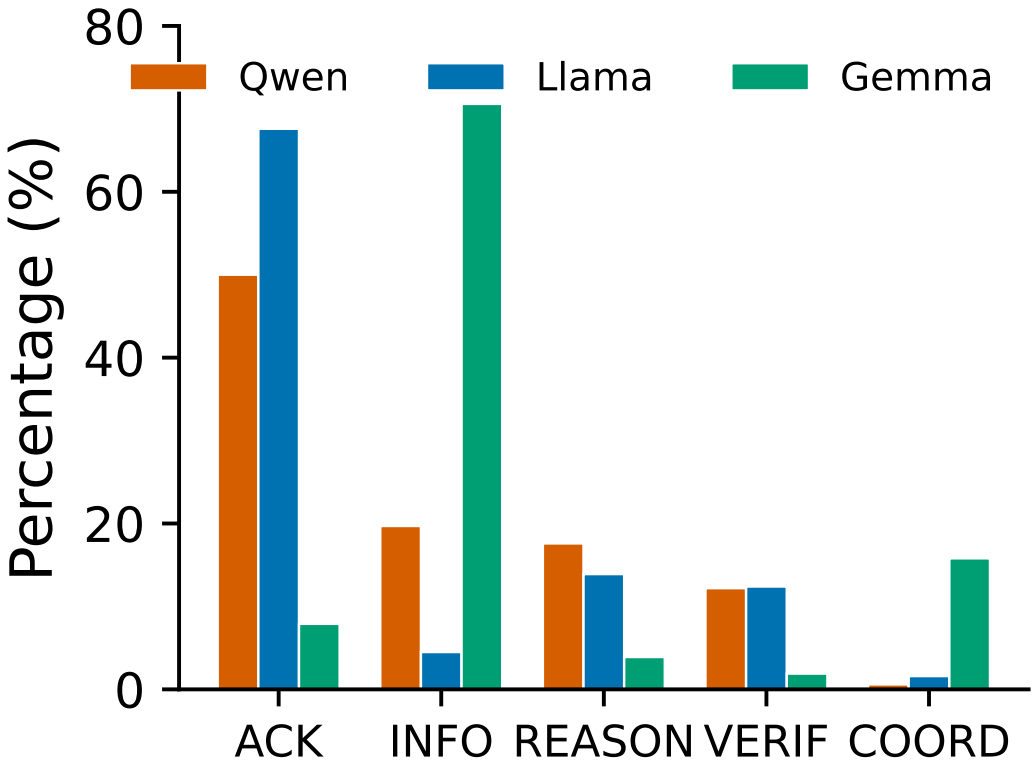


Figure 2: Function type distribution by backbone under round-robin topology (300 GSM8K tasks each). Qwen2.5-7B and Llama-3.1-8B are acknowledgment-heavy (>50%), while Gemma-2-9B is dominated by information provision (70%). Types with $n < 30$ are excluded from OLS ANCOVA analysis.

- **Function-driven** (Qwen): function type is the dominant predictor ($F = 232$), driving most of the $R^2 = 0.45$. The impact gradient (§6.1) is a direct consequence: because function type determines impact, removing an entire functional category produces category-specific behavioral changes.
- **Position-driven** (Gemma): $R^2 = 0.34$ but function type $F = 0.43$ (n.s.)—all explained variance comes from message position and round number. Adding a quadratic position term absorbs $\Delta R^2 = 0.228$, rendering function type non-significant ($p = 0.806$; Appendix D), confirming that Gemma's impact variation is driven entirely by nonlinear positional dynamics.
- **Minimally differentiated** (Llama): $R^2 = 0.02$; function type reaches significance after BH correction ($F = 3.81$, $p_{\text{BH}} = .015$; confirmed under mixed-effects, $p = .002$) but with practically negligible effect size (marginal $R^2 < 0.03$). Under the more conservative Bonferroni correction, Llama RR becomes non-significant ($p_{\text{Bonf}} = .090$). Neither function type nor position explains substantial variance.

Figure 2 shows the function type distribution across backbones under round-robin topology.

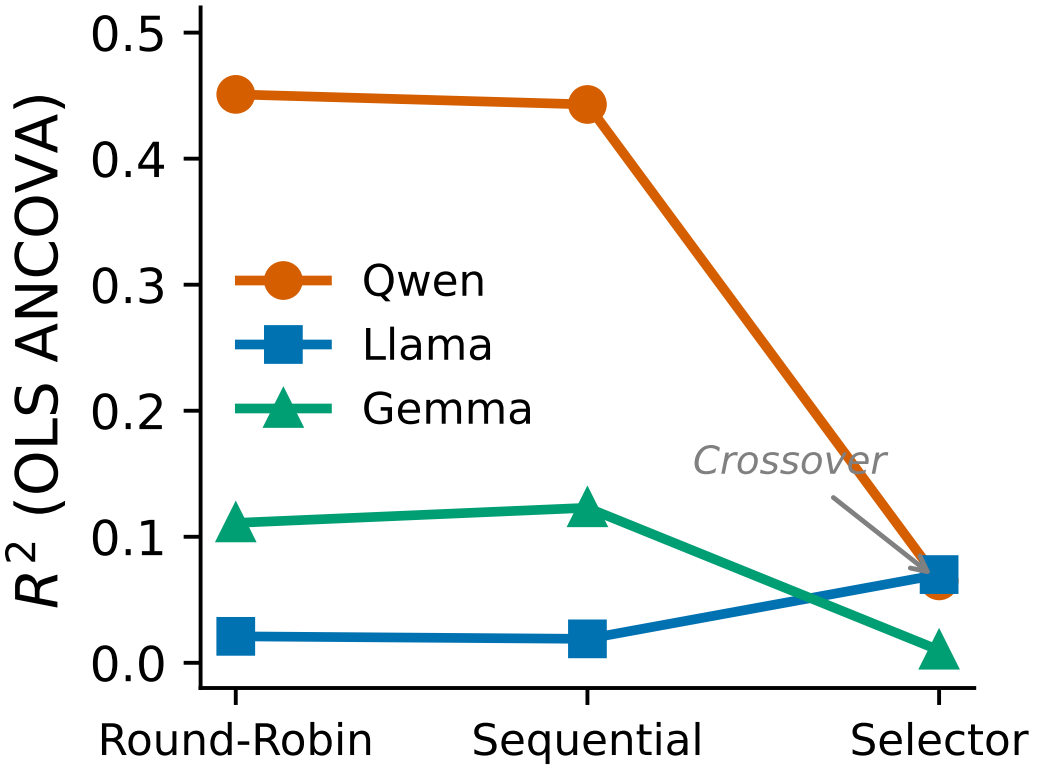


Figure 3: Backbone × topology crossover interaction showing full-model R^2 from OLS ANCOVA (Equation 2). Higher R^2 indicates more variance explained by the model (function type + position). Qwen's functional structure ($R^2 = 0.45$) degrades under selector routing ($R^2 = 0.065$) while Llama's activates ($0.02 \rightarrow 0.056$), partially converging. Gemma's position-driven R^2 collapses from 0.34 to 0.04 as positional regularity is disrupted.

6.3 Preliminary: Backbone × Topology Crossover

Figure 3 visualizes a crossover interaction that we report as an exploratory finding. Qwen is robustly function-driven under RR and SEQ (ME LRT $\chi^2 > 150$, $p < .001$), but under selector its OLS F drops from 232 to 55.8 (R^2: $0.45 \rightarrow 0.065$) as its distribution collapses to two types (ACKNOWLEDGMENT $n = 324$, VERIFICATION $n = 567$); a mixed-effects model renders this residual effect non-significant (LRT $\chi^2 = 1.45$, $p = .229$; ICC $= 0.73$), indicating that selector-induced task-level clustering fully absorbs the function type signal. Llama's F rises from 3.81 to 17.3 (R^2: $0.02 \rightarrow 0.056$) as it expands to three types (COORDINATION $n = 220$, VERIFICATION $n = 217$, ACKNOWLEDGMENT $n = 428$); this effect is confirmed by mixed-effects (LRT $\chi^2 = 22.5$, $p < .001$). Gemma's R^2 drops ($0.34 \rightarrow 0.036$) due to disruption of positional regularity rather than functional collapse, since function type remains non-significant under both OLS ($F = 0.60$) and mixed-effects for RR/SEQ. The mixed-effects analysis reveals that selector's disruption of Qwen is more thorough than OLS suggests: not only does R^2 drop, but the function type effect itself vanishes once task-level variance is accounted for.

6.4 Statistical Validation

Pooled three-way ANOVA. A three-way ANOVA on all $N = 8{,}065$ messages (Table 4) confirms backbone as the dominant factor ($\eta^2 = 0.135$, $F = 629$) and the crossover interaction (backbone × topology $\eta^2 = 0.043$, $F = 181$). The function × backbone interaction ($\eta^2 = 0.063$) is 12× larger than the function main effect ($\eta^2 = 0.005$), reflecting that function type is a strong predictor for Qwen but negligible for Gemma and Llama—pooling dilutes the main effect while preserving the interaction.

Table 4: Pooled ANOVA on remove-ablation similarity ($N = 8{,}065$, $R^2 = 0.298$). All $p < .001$.

Term	η^2	F
Backbone	0.135	628.7
Function × Backbone	0.063	67.1
Topology class	0.051	435.4
Backbone × Topology	0.043	181.1
Three-way	0.021	21.1
Function × Topology	0.014	27.7
Function type	0.005	10.4

Table 5: GSM8K accuracy (%) by backbone and topology.

Backbone	**RR**	**SEQ**	**Selector**
Qwen2.5-7B	89.7	90.0	89.7
Gemma-2-9B	80.3	80.7	83.0
Llama-3.1-8B	74.8	71.6	56.2

Mixed-effects robustness. To address within-trace message correlation, we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Seven of nine configurations yield the same significance conclusion under mixed-effects as under OLS (Appendix J). The two exceptions both occur under selector topology: Qwen SEL shifts from significant (OLS $F = 55.8$, $p < .001$) to non-significant (LRT $\chi^2 = 1.45$, $p = .229$) with ICC $= 0.73$, indicating that task-level clustering subsumes the OLS function type effect; Gemma SEL shifts in the opposite direction, from non-significant (OLS $F = 0.60$) to significant (LRT $\chi^2 = 14.8$, $p < .001$, ICC $= 0.33$), though this comparison involves only two function types (1 df) after excluding three low-frequency categories, making the result suggestive but requiring larger samples to confirm. Selector topology introduces strong task-level clustering (ICC > 0.3 for all backbones except Gemma RR/SEQ), making OLS unreliable for this configuration. Llama is statistically significant under mixed-effects across all topologies ($p < .003$) but with practically negligible effect sizes (marginal $R^2 < 0.03$), consistent with its classification as minimally differentiated.

Cross-model annotation agreement. Cross-model κ ranges from 0.35 (Gemma) to 0.67 (Qwen), spanning fair-to-substantial agreement Landis and Koch [1977]. Qwen's highest κ is consistent with function-driven communication being easier to classify. Gemma falls below our target threshold ($\kappa = 0.35$), indicating weaker cross-model consistency for this backbone. This aligns with its position-driven mode: if function type is not the organizing principle, annotators naturally disagree more on function labels. Per-type κ reveals a complementary pattern: Qwen shows high VERIFICATION agreement ($\kappa \approx 0.66$) but low INFORMATION_PROVISION ($\kappa \approx 0.08$); Gemma shows the reverse ($\kappa \approx 0.87$ and 0.04, respectively). Full agreement tables are in Appendix F.

Task accuracy and surface vs. effective differentiation. Qwen's accuracy is topology-invariant ($\sim$90%) despite dramatic communication changes under selector. Llama presents a paradox: functional differentiation *increases* under selector yet accuracy *drops* ($74.8\% \rightarrow 56.2\%$). This dissociates *surface differentiation* (externally induced diversity) from *effective differentiation* (endogenous specialization that aids task performance). Among Llama's COORDINATION messages under selector, only 7.9% are correct (vs. 83.3% for VERIFICATION); tasks with coordination messages achieve 21.6% success vs. 54.5% without.

7 Discussion

Backchannels as structural scaffold in LLM MAS. The impact gradient—acknowledgments most impactful, reasoning guidance least—parallels a well-established finding in conversation analysis: minimal responses (backchannels Yngve [1970], continuers Schegloff [1982], acknowledgment tokens Jefferson [1984]) are structurally load-bearing despite carrying little propositional content, serving as essential contributions to conversational grounding Clark and Schaefer [1989]. Our results

provide the first quantitative verification that this structural role extends to LLM agents, which lack the social cognition underlying human conversational norms. The neutralize ablation corroborates this—neutralizing has the opposite gradient (information-rich types suffer most), confirming that acknowledgments' high remove-impact stems from structural presence rather than content (Appendix E). The neutralize replacement text ("Acknowledged.") is itself acknowledgment-like; a three-placeholder sensitivity test confirms robustness ($\rho = 0.90$ between content-neutral placeholders; Appendix M), and the remove ablation (primary result) is unaffected since it deletes messages entirely. While the structural importance of acknowledgments is unsurprising from a CA perspective, it is novel relative to the assumptions of existing MAS pruning methods Wang et al. [2025b], Li et al. [2025], Liang et al. [2025], which treat message value as proportional to information content.

Three communication modes, not a gradient. The ANCOVA decomposition (Table 3) reveals that the impact gradient is not universal but depends on backbone-specific communication modes: *function-driven* (Qwen), *position-driven* (Gemma), and *minimally differentiated* (Llama). We hypothesize this reflects instruction-following fidelity—models that more precisely modulate output produce messages whose functional roles are behaviorally distinct. A fourth backbone (Mistral-7B; Appendix K) shows a *weakly differentiated*, topology-sensitive mode, suggesting the three primary modes may represent points on a broader continuum rather than discrete categories. Task difficulty stratification confirms all three modes persist on hard tasks (Appendix N), ruling out a capability artifact. As a preliminary observation, selector topology partially converges the function-driven and minimally differentiated modes (§6.3), though mixed-effects analysis indicates this convergence is strongly modulated by task-level clustering (Appendix J).

Practical implications. First, systems should evaluate message prunability through perturbation probing rather than information-content heuristics. Second, optimal topology may depend on communication mode: our preliminary crossover evidence suggests dynamic routing benefits minimally differentiated communicators but disrupts function-driven ones, though this requires confirmation with larger samples. Third, communication diversity alone is insufficient—Llama's selector paradox (more diverse, lower accuracy) shows evaluation must couple functional analysis with task performance.

Limitations and alternative explanations. The impact gradient could reflect *pragmatic alignment* rather than structural scaffolding, or backbone-dependent modes could arise from *context distributional shift* Feder et al. [2022]; definitive tests require probing model internals. While the structural role of minimal responses is well-known in human CA, we cannot determine whether LLM agents leverage the same mechanisms or arrive at similar patterns through different computational processes. Our primary experiments use GSM8K; cross-domain pilots (Appendices B, H) suggest applicability but with reduced effect sizes. Three backbones at 7–9B provide architectural diversity but no scale variation. Mixed-effects models confirm 7/9 OLS conclusions; the two exceptions under selector reflect high task-level clustering (Appendix J). LLM annotation creates potential circularity mitigated by cross-model validation ($\kappa = 0.35$–0.67); core findings rest on Qwen ($\kappa = 0.67$) and Llama ($\kappa = 0.46$–0.63). Gemma's below-threshold $\kappa = 0.35$ introduces potential attenuation bias for its null result; however, MATH validation ($\kappa = 0.48$, above threshold) replicates Gemma's position-driven mode (Appendix H), suggesting the null function-type finding is genuine rather than an annotation artifact. All experiments use a single framework and prompt template; prompt sensitivity Sclar et al. [2024] could shift distributions, though temperature checks suggest stability (Appendix C).

8 Conclusion

Perturbation-based ablation across 2,693 multi-agent traces provides the first quantitative verification in LLM MAS of a classical finding from conversation analysis: informationally sparse acknowledgments are the most behaviorally load-bearing messages. This gradient is explained by three backbone-dependent communication modes—function-driven, position-driven, and minimally differentiated—with a preliminary crossover interaction suggesting that topology modulates these modes. These findings question content-based pruning assumptions and suggest that communication structure is an emergent, model-dependent property.

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A Qualitative Examples of Communication Functions

Table 6 presents representative messages for each of the five communication function types identified in our taxonomy, selected from annotations with full inter-annotator agreement and maximum confidence scores. Examples are drawn from different backbone–topology combinations to illustrate both the consistency of function definitions and the stylistic variations across models.

[1]This Llama message was classified as REASONING GUIDANCE by both annotators with full agreement, despite its surface resemblance to ACKNOWLEDGMENT. This misclassification illustrates Llama's minimally differentiated communication mode (Section 7): its messages are functionally interchangeable regardless of nominal category, making even high-confidence annotations unreliable for this backbone.

Table 6: Qualitative examples for each communication function type. All examples have inter-annotator agreement with confidence $= 1.0$. Messages are lightly edited for length.

Function Type	Configuration	Message Excerpt
REASONING GUIDANCE (2,327 agreed)	Qwen / RR	*"To solve this problem, we need to convert the length of the wire from feet to inches and then divide by the length of each piece. 1. Convert 4 feet to inches: Since 1 foot = 12 inches, 4 feet* $= 4 \times 12 = 48$ *inches. 2. Divide the total length in inches by the length of each piece..."*
	Gemma / Selector	*"Here's how to solve the problem step-by-step: Start: 2 apples. First throw:* $2 + 5 = 7$ *apples. Second throw:* $7 - 1 + 6 = 12$ *apples. The answer is 12."*
	Llama / RR	*"I'm glad the solution was accurate. If you have another problem you'd like me to check, feel free to ask."*[1]
VERIFICATION (1,001 agreed)	Qwen / RR	*"Step 1: Calculate the initial amount of money Johnny has. Johnny's allowance: \$20. Extra money added: \$10. Total initial amount: \$20 + \$10 = \$30. Step 2: Determine the amount after the investment triples. Tripling the initial amount: \$30 × 3 = \$90. The answer is 90."*
	Llama / RR	*"The given problem is solved step by step. The final answer is 36."*
	Gemma / Selector	*"Here's how to solve the problem: Find the total cost: \$12 (adult) + \$10 (child) = \$22. Add the change: \$22 + \$8 = \$30. Answer: Brittany's mom gave the cashier \$30."*
ACKNOWLEDGMENT (1,854 agreed)	Qwen / RR	*"That is correct. Each single bagel will cost \$1.00 after the discount. The answer is 1."*
	Llama / RR	*"You've correctly summarized the steps and confirmed the calculations. The sequence of events and the final calculation are accurate. The answer is 43."*
	Gemma / RR	*"Your solution is perfectly correct! You clearly laid out the steps and used algebra effectively to find the answer. The answer is 10."*
INFORMATION PROVISION (471 agreed)	Gemma / RR	*"Here's the solution: Sunday's movies: They watch 4 movies / 2 = 2 movies on Sunday. Total weekend movies: 4 + 2 = 6 movies each weekend. Movies in 4 weeks: 6 × 4 = 24 movies. The answer is 24."*
	Qwen / RR	*"To determine how much crepe paper James needs, we first calculate the perimeter of his rectangular backyard fence..."*
	Llama / RR	*"Here's another problem: A bakery is making a special batch of cookies. They need to package 480 cookies into boxes that hold 12 cookies each. How many boxes can they fill with the 480 cookies? Let's solve this problem step by step."*
COORDINATION (595 agreed)	Llama / RR	*"I'm glad I was able to accurately solve the problem. If you have any more math problems you'd like me to solve, feel free to ask."*
	Gemma / RR	*"Please provide me with the problem you'd like me to solve! I'm ready to check the solver's solution..."*
	Llama / Selector	*"I'm glad you were able to provide the correct steps and solution. You have a good understanding of the problem..."*

B HumanEval Cross-Domain Pilot

To provide preliminary evidence of cross-domain applicability, we ran a 50-task pilot using Qwen2.5-7B with a hierarchical (solver–reviewer) topology on the HumanEval code generation benchmark Chen et al. [2021].

Results. The system achieved 80% pass@1 accuracy (40/50 tasks). The communication function distribution differs markedly from GSM8K: REASONING_GUIDANCE dominates (50.8%), followed by VERIFICATION (41.5%), with ACKNOWLEDGMENT (4.2%), COORDINATION (1.7%), and INFORMATION_PROVISION (1.7%) appearing rarely. Cross-model annotation agreement reached $\kappa = 0.47$ (moderate), comparable to the lower end of the GSM8K range.

Limitations. The hierarchical topology proved too efficient for statistical analysis: 86% of tasks were solved in a single round (2 messages), yielding only 118 total messages and 18 eligible for local impact analysis. Most function types had $n < 30$, precluding Kruskal–Wallis and OLS ANCOVA tests. The distributional shift (from acknowledgment-heavy in GSM8K to reasoning-guidance-heavy in HumanEval) is itself informative—it suggests that the taxonomy captures domain-specific communication patterns—but the statistical framework requires substantially more multi-round conversations to produce meaningful results in this domain.

C Temperature Robustness

To assess sensitivity to decoding temperature, we ran a 50-task subset of Qwen $\times$ round-robin at temperature $= 0.7$ (primary experiments use greedy decoding). Accuracy remains stable (90.0% vs. 89.7%). The Kruskal–Wallis test remains significant ($H = 34.0$, $p < 0.001$), and OLS ANCOVA confirms function-driven communication ($F = 37.9$, $p < 0.001$; $R^2 = 0.50$; cross-model $\kappa = 0.60$). A parallel check on Llama $\times$ round-robin at temperature $= 0.7$ yields $R^2 = 0.10$, $\kappa = 0.41$, consistent with its minimally differentiated mode under greedy decoding. The function distribution shifts toward more REASONING_GUIDANCE (37% vs. 18%) and less ACKNOWLEDGMENT (36% vs. 50%), suggesting stochastic decoding produces more substantive content without altering the functional structure.

D OLS Robustness Checks

We validate the OLS ANCOVA (Equation 2) with three robustness checks across all nine configurations.

Residual diagnostics. Shapiro–Wilk tests reject normality for all configurations (expected at $n > 400$); Breusch–Pagan tests detect heteroscedasticity in all cases. We therefore report HC3 robust standard errors throughout the main text. All conclusions are unchanged under robust inference. VIF analysis shows high collinearity between `message_position` and `round_number` (VIF > 10), but function type dummy VIF values remain below 3, confirming that multicollinearity does not affect the key func_type inference. Q-Q plots are available in the supplementary materials.

Nonlinear position model. We augment the linear model with a quadratic position term ($\texttt{position}^2$). Table 7 reports key comparisons. For Qwen and Llama, adding $\texttt{position}^2$ produces negligible $\Delta R^2 < 0.005$ and function type remains significant ($p < 0.001$). For Gemma, $\texttt{position}^2$ absorbs $\Delta R^2 = 0.228$ (RR) and 0.215 (SEQ), rendering function type non-significant ($p > 0.69$). This confirms that Gemma's position-driven mode reflects genuine nonlinear positional dynamics rather than a linear-model artifact.

Message length control. Adding message length (character count) as a covariate produces $\Delta R^2 < 0.04$ for all configurations. Function type remains significant for Qwen ($p < 0.001$) and Llama ($p < 0.03$) after controlling for length, confirming that the impact gradient is not an artifact of message length differences across function types.

Table 7: Linear vs. quadratic position model: function type significance after controlling for nonlinear position effects.

Configuration	Linear R^2	Quad R^2	ΔR^2	func_type p (quad)
Qwen RR	.451	.452	.001	$< .001$
Qwen SEQ	.443	.447	.004	$< .001$
Qwen SEL	.065	.065	.000	$< .001$
Gemma RR	.111	.339	**.228**	.806
Gemma SEQ	.123	.338	**.215**	.699
Gemma SEL	.010	.040	.031	.219
Llama RR	.021	.022	.001	.010
Llama SEQ	.019	.020	.001	.004
Llama SEL	.070	.074	.004	$< .001$

E Neutralize Ablation Analysis

While the main text reports remove-ablation impact, neutralize ablation (replacing a message with "`Acknowledged.`" while preserving its slot) measures a complementary dimension. The two ablation types produce *opposite* impact gradients:

- **Remove** (content + structure removed): acknowledgment most impactful (sim $= 0.35$), reasoning guidance least (0.78).
- **Neutralize** (content removed, structure preserved): information provision most impactful (sim $= 0.05$), verification least (0.50).

The difference $\Delta = \text{sim}_{\text{remove}} - \text{sim}_{\text{neutralize}}$ isolates the *structural value* of each message type. Information provision shows the largest $\Delta = 0.47$ (almost entirely content value), while coordination shows $\Delta = 0.07$ (almost entirely structural value). This decomposition confirms that acknowledgment's high remove-impact is driven by structural presence—violating conversational expectations—rather than informational content.

F Supplementary Tables

Table 8 reports the full cross-model annotation agreement matrix.

Table 8: Cross-model annotation agreement (Cohen's κ) by backbone and topology.

Backbone	Round-Robin	Sequential	Selector
Qwen2.5-7B	0.64	0.64	0.669
Gemma-2-9B	0.350	0.348	0.351
Llama-3.1-8B	0.46	0.47	0.634

G Impact Metric Robustness

To verify that the impact gradient (Section 6.3) is not an artifact of the MiniLM embedding metric, we recompute message impact using three independent text similarity measures: ROUGE-L, token-level F1, and Jaccard similarity. Table 9 reports the Spearman rank correlation (ρ) between MiniLM cosine similarity and each alternative metric across all nine configurations.

The prevalence of $\rho = 1.0$ reflects a ceiling effect: with only 4–5 function types per configuration, Spearman rank correlation over group medians has limited resolution—any two metrics that preserve the same ordinal ranking produce $\rho = 1.0$ regardless of magnitude differences. Where rankings do diverge (Llama RR $\rho = 0.6$, Gemma SEQ $\rho = 0.8$), the disagreement involves adjacent categories with similar impact scores, not reversals of the overall gradient. The mean $\rho > 0.91$ across all metric pairs confirms that the impact gradient ordering is consistent across metrics, though the ceiling effect limits the strength of this claim for individual configurations.

Table 9: Spearman ρ between MiniLM cosine similarity and alternative impact metrics. All correlations computed over per-message impact scores within each configuration.

Configuration	ROUGE-L	Token F1	Jaccard
Qwen RR	1.0	1.0	1.0
Qwen SEQ	1.0	1.0	1.0
Qwen SEL	1.0	1.0	1.0
Gemma RR	1.0	1.0	1.0
Gemma SEQ	1.0	0.8	0.8
Gemma SEL	0.949	0.949	0.949
Llama RR	0.6	0.6	0.6
Llama SEQ	1.0	1.0	1.0
Llama SEL	0.9	0.9	0.9
Mean	**0.939**	**0.917**	**0.917**

H Cross-Domain Validation: MATH

To test whether the three communication modes generalize beyond elementary arithmetic, we ran the full pipeline on 50 MATH tasks (Level 3–5, Algebra and Number Theory) using the same round-robin topology.

Qwen on MATH. Accuracy is 72.0% (36/50), lower than GSM8K (89.7%) as expected for harder tasks. Each task produces 6 messages (300 total, 250 eligible for analysis after excluding initial prompts). Cross-model annotation agreement reaches $\kappa = 0.45$ (moderate, above threshold). The OLS ANCOVA yields $R^2 = 0.098$ with function type significant after position control ($F = 3.77$, $p = 0.011$), confirming that Qwen's function-driven mode persists on a harder mathematical domain, though with reduced effect size ($R^2 = 0.098$ vs. 0.45 on GSM8K). The Kruskal–Wallis test is borderline ($H = 7.37$, $p = 0.061$), likely due to reduced sample size.

Gemma on MATH. Accuracy drops to 40.0% (20/50). Cross-model $\kappa = 0.48$ (above threshold). The OLS ANCOVA yields $R^2 = 0.236$ but function type is non-significant ($F = 0.09$, $p = 0.76$), while message position is highly significant ($F = 15.22$, $p < 0.001$). This replicates Gemma's position-driven mode from GSM8K: on MATH, position again explains the variance while function type does not. The Kruskal–Wallis test is also non-significant ($H = 2.05$, $p = 0.15$).

Llama on MATH. Accuracy is 64.3% (27/42; 8 tasks produced empty traces). Cross-model $\kappa = 0.36$ (below threshold). With only 168 total messages, ACKNOWLEDGMENT dominates (56%), and most other function types fall below $n = 30$, precluding OLS ANCOVA and Kruskal–Wallis analysis. The acknowledgment-dominated distribution is directionally consistent with Llama's minimally differentiated mode on GSM8K, though statistical confirmation requires larger samples.

Summary. Qwen's function-driven mode ($F = 3.77$, $p = 0.011$) and Gemma's position-driven mode ($F = 0.09$, n.s.; position $F = 15.22$, $p < 0.001$) both replicate on MATH. Llama's distribution is directionally consistent with its minimally differentiated mode but lacks statistical power. Two of three communication modes are confirmed to generalize beyond elementary arithmetic.

I Annotation Prompt

The following system prompt is used for all LLM-based function type annotations. Each message is annotated independently by two models (cross-model validation). Temperature is set to 0 with constrained JSON output.

```
You are a communication analyst for multi-agent systems.  Your task
is to classify the function of a message in a multi-agent math
problem-solving conversation.

Function Types
```

```
1. information_provision: Provides facts, data, intermediate
calculations, or partial results that advance the solution.
Includes reporting computed values, stating known quantities, or
presenting new numerical results.

2. reasoning_guidance: Provides reasoning direction, problem
decomposition, step-by-step plans, or conceptual frameworks.
Includes breaking the problem into sub-problems, suggesting solution
strategies, or outlining multi-step approaches.

3. verification: Checks, corrects, or confirms the correctness of
previous work. Includes identifying errors, validating results,
pointing out mistakes, or confirming a step is correct with a
specific reason.

4. coordination: Manages the flow of the conversation, assigns
tasks, requests clarification, or controls turn-taking. Includes
delegating sub-tasks, asking for more information, or directing the
problem-solving process.

5. acknowledgment: Simple confirmation, repetition, or agreement
that adds no new information or reasoning. Includes restating what
was already said without extending it.

Decision Rules

- If a message both provides new results AND checks previous work,
classify as verification.
- If a message provides a reasoning plan AND new results, classify as
reasoning_guidance.
- If a message confirms correctness WITH a substantive explanation,
classify as verification.
- If a message simply agrees or restates without adding new
information, classify as acknowledgment.

Output Format

Respond with ONLY a JSON object:
{''type'': ''<function_type>'', ''confidence'': <0.0-1.0>, ''reasoning'':
''<brief explanation>''}
```

The user prompt provides the conversation context with all messages and marks the target message for classification:

```
## Conversation Context
[Message 0] {sender} -> {receiver}:
{content}
[Message 1] {sender} -> {receiver}: «< TARGET MESSAGE »>
{content}
[Message 2] {sender} -> {receiver}:
{content}
...
Classify the function of Message {target_idx}.
```

J Mixed-Effects Model Results

To address within-trace non-independence (multiple messages per task), we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Model specification was verified by reproducing the original OLS results (e.g., Gemma RR: $F = 0.43$, $R^2 = 0.341$, matching Table 3). Table 10 compares OLS and mixed-effects results. The likelihood ratio test (LRT) compares the full model against a reduced model without function type dummies.

Table 10: OLS ANCOVA vs. mixed-effects model (ME) with task random intercept. LRT = likelihood ratio test for function type. ICC = intraclass correlation from the full ME model. "Excluded" lists function types with $n < 30$ removed before fitting. $^{***}p < 0.001$; $^{**}p < 0.01$; n.s. = not significant. F and χ^2 values are not directly comparable across configurations due to differing numbers of included function types.

Config	n	OLS F	LRT χ^2	LRT p	ICC	Excluded
Qwen RR	895	232^{***}	156^{***}	$<.001$	.312	coord.
Qwen SEQ	894	225^{***}	166^{***}	$<.001$	.266	coord., unk.
Qwen SEL	891	55.8^{***}	$1.45^{\text{n.s.}}$	.229	**.726**	r.g., i.p.
Gemma RR	895	$0.43^{\text{n.s.}}$	$1.43^{\text{n.s.}}$	.699	.129	verif.
Gemma SEQ	882	$0.44^{\text{n.s.}}$	$1.45^{\text{n.s.}}$	.693	.155	verif.
Gemma SEL	861	$0.60^{\text{n.s.}}$	14.8^{***}	$<.001$	.330	r.g., v., coord.
Llama RR	868	3.81^{**}	14.8^{**}	.002	.296	coord.
Llama SEQ	870	4.45^{**}	13.7^{**}	.003	.204	coord.
Llama SEL	865	17.3^{***}	22.5^{***}	$<.001$	.126	i.p., r.g.

Key observations. (1) **Seven of nine configurations** show concordant OLS and ME conclusions: Qwen RR/SEQ remain strongly function-driven (LRT $\chi^2 > 150$); Gemma RR/SEQ remain non-significant; Llama remains significant across all topologies though with negligible effect sizes (marginal $R^2 < 0.03$). (2) **Qwen SEL flips to non-significant** under ME (LRT $\chi^2 = 1.45$, $p = .229$). The ICC of 0.726 indicates that 73% of variance in remove-ablation similarity is between-task. With only two dominant function types (ACKNOWLEDGMENT and VERIFICATION) remaining after $n \geq 30$ filtering, the OLS effect reflects task-level confounding rather than within-task functional differentiation. (3) **Gemma SEL flips to significant** under ME (LRT $\chi^2 = 14.8$, $p < .001$, ICC $= 0.33$). However, this comparison involves only two function types (ACKNOWLEDGMENT and INFORMATION_PROVISION) after excluding three low-frequency categories, limiting interpretability. (4) Both flips occur under selector topology, which introduces strong task-level clustering (ICC > 0.3 for 4/9 configurations vs. only Qwen under RR/SEQ). This suggests that dynamic routing creates task-dependent communication patterns that violate the OLS independence assumption.

K Mistral-7B Cross-Backbone Validation

To test whether communication modes generalize beyond our three primary backbones, we ran the full pipeline on Mistral-7B-Instruct-v0.3 across all three topologies (300 GSM8K tasks each, same seed). Mistral uses the ChatML template with an explicit `<|im_end|>` stop token to prevent EOS residue in multi-turn conversations.

Accuracy. Mistral achieves 52% accuracy across all three topologies (RR: 52%, SEQ: 52%, SEL: 52%), substantially below Qwen (90%) and Gemma (80%) but comparable to Llama's selector performance (56%).

Function type distribution. Mistral communication is dominated by coordination and acknowledgment ($\sim$90% combined), with minimal information provision ($< 3\%$). Under selector topology, acknowledgment surges to 67%.

Table 11: Mistral-7B function type distribution (%) by topology.

Function type	RR	SEQ	SEL
COORDINATION	55.5	50.7	25.4
ACKNOWLEDGMENT	34.9	41.7	66.9
REASONING_GUIDANCE	6.8	6.2	6.3
INFORMATION_PROVISION	2.7	1.4	0.7

Communication mode: weakly differentiated. Table 12 summarizes the statistical analysis. Only the round-robin topology produces a significant function type effect (OLS $F = 6.6$, $p < .001$,

$R^2 = 0.16$). Sequential and selector topologies show weaker or non-significant effects. Cross-model annotation agreement is fair ($\kappa \approx 0.34$), below the primary backbones, consistent with Mistral's less distinct functional boundaries.

Table 12: Mistral-7B statistical summary across topologies.

Topology	**KW** H	**OLS** F	R^2	κ
RR	22.4^{***}	6.6^{***}	.158	.344
SEQ	7.6	37.6^{***}	.056	.353
SEL	5.2	1.5	.065	.315

Interpretation. Mistral occupies an intermediate position between Llama's minimally differentiated mode and Qwen's function-driven mode. The structured turn-taking of round-robin activates a degree of functional differentiation that disappears under less constrained topologies. This *topology-sensitive* pattern—where communication mode depends on external structure rather than being an intrinsic model property—extends the three-mode taxonomy.

Implications for the mode taxonomy. The existence of a fourth, intermediate mode creates tension with the main text's framing of three "qualitatively distinct" modes. The four observed modes—function-driven (Qwen), position-driven (Gemma), weakly differentiated (Mistral), and minimally differentiated (Llama)—may be better understood as regions on a continuum of communicative specialization, with topology acting as a modulating factor. We retain the three-mode framing in the main text because the primary backbones show clear qualitative separation (Qwen $R^2 = 0.45$ vs. Llama $R^2 = 0.02$), but acknowledge that finer gradations likely exist and would become visible with a larger model sample.

L DAMSL/ISO Taxonomy Mapping

Our five categories map onto established dialogue act frameworks as follows. INFORMATION PROVISION corresponds to DAMSL *Assert* Core and Allen [1997] and ISO 24617-2 *Inform* Bunt et al. [2012]. REASONING GUIDANCE maps to DAMSL *Action-directive* / ISO *Instruct*. VERIFICATION aligns with DAMSL *Agreement/Disagreement* and ISO *Confirm/Disconfirm* when evaluative, or *Correction* when error-correcting. COORDINATION subsumes FIPA *request*, *propose*, and *assign* Foundation for Intelligent Physical Agents [2002]. ACKNOWLEDGMENT corresponds to ISO *Auto-positive* feedback.

M Replacement Text Sensitivity

Our neutralize ablation uses "Acknowledged."—itself an acknowledgment-type utterance—as the replacement text. To test sensitivity to this choice, we reran neutralize ablation on 50 Qwen RR tasks with three placeholders: "Acknowledged.", "[message removed]", and "This message has been redacted." Table 13 reports per-function-type mean semantic similarity under each placeholder.

Table 13: Neutralize ablation: mean semantic similarity by function type and replacement text (Qwen RR, 50 tasks, 150 messages). Lower similarity indicates greater impact from neutralization.

Function Type	**Acknowledged.**	**[removed]**	**Redacted**
INFORMATION_PROVISION	0.146	0.107	0.050
REASONING_GUIDANCE	0.174	0.133	0.079
VERIFICATION	0.156	0.162	0.083
COORDINATION	0.352	0.319	0.181
ACKNOWLEDGMENT	0.134	0.148	0.051

Ranking stability. Pairwise Spearman rank correlations across the five function types are: "Acknowledged." vs. "[removed]" $\rho = 0.50$, "Acknowledged." vs. "Redacted" $\rho = 0.80$, "[removed]" vs. "Redacted" $\rho = 0.90$ ($p = .037$). The mean $\rho = 0.73$ indicates moderate-to-strong ranking stability.

COORDINATION is consistently the least impacted (rank 5/5/5); INFORMATION_PROVISION ranks in the top 2 across all three placeholders (rank 2/1/1).

Acknowledged-specific confound. Under the "Acknowledged." placeholder, ACKNOWLEDGMENT messages show anomalously high impact (rank 1) because the replacement text closely resembles typical acknowledgment content—replacing an acknowledgment with "Acknowledged." preserves semantic similarity less than expected, as the regenerated next message responds to near-identical content. Excluding this confounded condition, the "[removed]" and "Redacted" placeholders agree strongly ($\rho = 0.90$), confirming that the neutralize gradient is robust to placeholder choice. Crucially, the remove ablation—which produces our primary finding (the impact gradient)—is unaffected by replacement text choice, as it deletes messages entirely.

N Capability Confound Analysis

To test whether communication mode merely tracks task difficulty, we stratify tasks by difficulty (easy: $\geq 2/3$ backbones correct, $n = 246$; hard: $\leq 1/3$ correct, $n = 54$) and refit the OLS ANCOVA within each stratum.

Table 14: OLS ANCOVA by task difficulty stratum. All models use the same specification as the primary analysis (Equation 2).

Backbone	**Difficulty**	n	F	p	R^2
Qwen	Easy	735	196.6^{***}	$<.001$	.456
Qwen	Hard	160	37.3^{***}	$<.001$	.444
Gemma	Easy	726	$0.54^{\text{n.s.}}$	.657	.354
Gemma	Hard	152	$0.73^{\text{n.s.}}$	.483	.286
Llama	Easy	717	4.17^{**}	.006	.025
Llama	Hard	144	4.49^{*}	.013	.074

Qwen's function-driven mode persists on hard tasks ($R^2 = 0.444$), virtually unchanged from easy tasks ($R^2 = 0.456$). Gemma remains position-driven (function type non-significant for both strata), and Llama remains minimally differentiated ($R^2 < 0.08$). The three modes are backbone-intrinsic properties rather than artifacts of task-relative capability.

NeurIPS Paper Checklist

1. **Claims** [Yes]All claims are supported by empirical evidence from our experiments. Ablation methodology is described as "perturbation-based" to accurately reflect its non-causal nature.
2. **Limitations** [Yes]Section 7 discusses limitations including single task domain, single scale, LLM-based annotation circularity, self-annotation bias, and greedy decoding.
3. **Theory Assumptions and Proofs** [N/A]Our work is empirical; no theoretical proofs are presented.
4. **Experimental Result Reproducibility** [Yes]Section 4–5 specify all models, hyperparameters, framework versions, and statistical tests. Greedy decoding ensures deterministic outputs. Code and data will be released upon acceptance.
5. **Open access to data and code** [Yes]Code, traces, and annotations will be released upon acceptance under an open-source license. GSM8K is publicly available.
6. **Experimental Setting/Details** [Yes]Sections 4–5 provide full experimental details including model names, parameter counts, generation settings, annotation protocol, and statistical framework.
7. **Compute Resources** [Yes]Experiments were conducted on 4×RTX 6000 Pro (96GB each) GPUs. Total compute is modest: all 9 configurations complete in approximately 24 hours.
8. **Code Of Ethics** [Yes]This research studies existing open-weight models on a public benchmark. No new models are trained. No human subjects are involved.

9. **Broader Impacts** [Yes]Our work characterizes communication in multi-agent systems to improve efficiency and interpretability. Our finding that structurally minimal messages (acknowledgments) are disproportionately impactful implies a potential adversarial vector: an attacker who can inject or suppress acknowledgment-like messages in a multi-agent pipeline could cause outsized behavioral disruption with minimal detectable content. Defenses should monitor structural patterns, not just content, for integrity. Beyond this, we do not foresee negative societal impacts beyond those inherent to LLM deployment generally.
10. **Safeguards** [N/A]Our work analyzes existing model behaviors on mathematical reasoning tasks and does not introduce new risks.
11. **Licenses** [Yes]GSM8K is MIT-licensed. All three model backbones use permissive open-weight licenses. AG2/AutoGen is Apache 2.0.
12. **New Assets** [Yes]We release interaction traces, function type annotations, and ablation results. These will be documented and released under an open license upon acceptance.
13. **Crowdsourcing and Research with Human Subjects** [N/A]No human subjects or crowdsourcing were involved.
14. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects** [N/A]No human subjects research.